

Asset Ambiguity, Reporting Flexibility, and Strategic Declarations: Evidence from Audit Announcements and a Dividend Tax Cut

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Abstract

This paper studies how ambiguity in asset classification and flexibility in tax reporting enable strategic tax declarations. Using administrative data from the Netherlands, we show that a 2005 audit announcement coincided with sharp increases in declared property-related assets, particularly among the wealthiest taxpayers, in sections not directly targeted by the announcement. The study also exploits a one-year dividend tax cut in 2007, which coincided with a second audit announcement, to further examine strategic spontaneous declarations and income shifting among shareholders, particularly those with substantial company holdings. The findings reveal how taxpayers exploit reporting flexibility to declare previously hidden and hard-to-trace wealth. The results underscore how classification ambiguity can be used to strategically reallocate wealth in response to tax policy changes.

Keywords: Tax Ambiguity, Reporting Flexibility, Tax Evasion, Auditing Announcements, Dividend Tax

JEL Codes: H26, H83, K34

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1 Introduction

A longstanding literature on tax evasion considers taxpayers’ static and dynamic responses to random audits and audit letter nudges (Slemrod et al., 2001; Kleven et al., 2011; Fellner et al., 2013; Slemrod, 2019). More recent work exploiting offshore account data leaks and tax amnesties shifts focus to taxes evaded by the wealthiest (Alstadsæter et al., 2019; Guyton et al., 2021; Leenders et al., 2023; Londoño-Vélez and Avila-Mahecha, 2024). This literature has grown in conjunction with that on behavioural responses to changes in dividend and wealth taxes (Chetty and Saez, 2006; Scheuer and Slemrod, 2021; Rotberg and Steinberg, 2024). Several well-known insights emerge from this research: tax evasion occurs most often on items not verified by third-party information; taxpayers react to announcements concerning newly held fiscal information; and the upper quantiles of the wealth distribution have more avenues for legally avoiding taxes and account for the largest amounts of evaded tax in absolute terms.

While much is known about how taxpayers respond to enforcement and information, less is understood about how ambiguity in asset classification and flexibility in reporting contribute to tax evasion and avoidance, particularly among the wealthy. This paper exploits a series of unusual audit announcements in the Netherlands to document declaration patterns consistent with strategic responses to scrutiny. Using administrative data, this study documents large jumps in the assets targeted by the announcements, but not in the sections of the tax return directly targeted by those announcements. It further exploits a one-year dividend tax cut, which coincided with one of the announcements, to more narrowly assess strategic spontaneous declarations and income shifting among shareholders, particularly those with substantial company holdings.

Each January until 2012, the Netherlands Tax and Customs Administration publicly announced a specific item from the previous year’s tax returns that would be subject to intensified auditing. These announcements - broadcast via radio, television, newspapers, and online media - were designed to raise salience and improve compliance. Two announcements, in 2005 and 2007, focused on particularly complex topics with broad implications.

The 2005 announcement targeted *income from side earnings*, a broad category reported in Box 1, one of three sections of the Dutch tax form. Declaring income in this category prompts taxpayers to complete a linked entry *on profits from assets made available*, also in Box 1. In turn, taxpayers must indicate whether these earnings relate to three additional categories: *other assets*, *other property*, and *debts*. These categories include, among other things, rental income (excluding primary and secondary residences), various dividends and capital gains, and financial liabilities such as mortgages.

Beyond their ambiguous definition, these three items are unique in that they can be declared in Box 1, the section of the Dutch tax form related to income; Box 3, the section related to wealth; and, in some circumstances, Box 2, which encompasses income from a substantial business interest. Above a relatively low monetary threshold, the tax rates in Box 3 are significantly lower than those in Boxes 1 and 2.

The paper uses of detailed administrative longitudinal data from the Netherlands Tax and Customs Administration covering 2002-2008. Because all taxpayers are exposed to the spot-light announcements and the dividend tax cut, we cannot credibly formulate separate treatment

and control groups. Accordingly, the results should be considered as lying somewhere between descriptive and causal, given that the proposed methodology does not use what would traditionally be considered a causal approach. However, the tax code’s structure and the nature of the incentive changes allow for clear ex ante predictions about which tax categories should be affected. We interpret the findings as quasi-experimental evidence, combining institutional variation with sharp, testable implications.

The first set of results shows that the 2005 announcement saw only small increases in declarations of *income from side earnings* and no changes in the linked Box 1 category *profits from assets made available*. It did, however, produce notable jumps in Box 3 declarations of *other property* and *other assets*.

The second set of results focuses on wealth changes in 2007 coincided with two concurrent events influencing tax incentives. The first was the 2007 auditing announcement, which focused on all items in Box 3, the tax section grouping returns for different categories of investments and savings. The second was a one-year drop in the dividend tax rate from 25% to 22% on the first €250,000 declared in Box 2, the section relating to income from a substantial business interest (over 5% shareholding in a company). The analysis does not indicate a reversal of movements from Box 3 for *other assets*, *other property* or *debts* back to Box 1. They do, however, show a jump in *profits from assets made available* in Box 1. In addition, the results show that the Box 2 rate cut led to a €1.9 billion one-year surge in declared capital income, the payouts of which appear mainly in Box 2, but also in Box 3 categories such as *share dividends and bonds*. This revelation of otherwise withheld company profits led to several policy changes over the following years, which we discuss below.¹

Together, the audit announcement and dividend tax cut highlight how taxpayers exploit both classification ambiguity and reporting flexibility to strategically declare income. They also reveal that responses to short-term incentives do not stay confined to a single tax category but often spill across multiple sections of the return. The two episodes complement one another: the 2005 announcement captures behavioral responses to changes in perceived audit risk, while the 2007 episode captures responses to a change in the marginal tax rate. The key distinction is that the 2005 declarations reflect previously hidden wealth, whereas the 2007 surge represents legal income shifting in response to tax incentives.

The paper further explores the intersection of the two policies by examining subsamples most likely to respond, focusing on heterogeneity by employment, filing behavior, and demographic characteristics. One notable finding is that the observed shifts are present for taxpayers filing with and without the aid of a tax professional, though they are substantially larger for the former group. The 2007 jump in Box 2 declarations is entirely driven by taxpayers who did not file as entrepreneurs, consistent with the intent of the dividend tax rate cut. Across socio-demographic groups, effects are concentrated among individuals with characteristics most strongly associated with high wealth, particularly those employed in the financial and business services sector.

This paper contributes to several strands of the literature. First, we add to research showing how announced increases in auditing in one tax area or for a set of taxpayers can produce spillovers to other taxes, over time, or across taxpayers (Le Maire and Schjerning, 2013; Devereux

¹The direct effect of the tax cut on Box 2 declarations has already been documented, among others, in Bettendorf et al. (2017).

et al., 2014; Boning et al., 2020). In particular López-Luzuriaga and Scartascini (2019) and Castro et al. (2022) show that various forms of auditing letter messages - penalties, detection, social norms - can increase declarations for the targeted tax item but also produce negative spillovers on other items if agents perceive auditing capacity as limited. Wealth shifting can also be motivated by purely financial reasons, as shown by Alstadsæter and Jacob (2016), who document that a permanent reduction in the dividend tax rate leads to income shifting across tax returns, particularly among director-large shareholders.

More broadly, this paper shows how taxpayers exploit both classification ambiguity and reporting flexibility to strategically declare income in response to enforcement and tax policy changes. Even seemingly small or narrowly targeted tax reforms can therefore generate important and unintended knock-on effects across different sections of the tax return. In the current context, ambiguity over how certain assets should be classified across tax categories creates additional scope for wealthy taxpayers to reallocate income and wealth in ways that reduce scrutiny. These channels compound inequalities in tax advantages arising from differences in resources - such as access to tax professionals - and in human capital, including the ability to navigate complex tax systems (Abeler and Jäger, 2015; Bhargava and Manoli, 2015; Feldman et al., 2016; Craig and Slemrod, 2024), and differences in access to offshore banks and other international wealth holdings (Alstadsæter et al., 2022b; Bomare and Herry, 2022; Johannesen et al., 2022). In addition, this paper offers insights into the effects of a one-time dividend tax cut, which may be different from those of a permanent dividend tax cut (Chetty and Saez, 2006; Alstadsæter et al., 2022b). Finally, it underlines how the ambiguity of certain assets' classification leaves discretion for wealthy taxpayers to re-optimize their tax declarations strategically in response to new tax policies.

2 Institutional Background

In this section, we first describe the Dutch tax system, then outline the institutional features of the annual audit announcements used by the Netherlands Tax and Customs Administration, including the procedures for their selection, public communication, and enforcement. We then describe the specific announcements that are central to our analysis, focusing in particular on the 2005 campaign targeting income from side earnings. Finally, we outline the circumstances leading to the 2007 dividend tax cut.

2.1 Dutch tax system

Since 2001, the Dutch tax system separates tax declarations into three categories, or Boxes, as described in Table 1. About 70% of the Dutch population files tax returns annually, and tax is levied on income net of any deductibles within each Box. Box 1 relates to wages, profits, social security benefits and pensions. It follows a progressive tax schedule with four brackets, which in 2005 had cutoffs at €16,893, €30,357 and €51,762. The income in each bracket is taxed at 1.80%, 9.35%, 42% and 52%. For income under €30,357 there is also a 32.60% flat rate for social security contributions. The second category, Box 2, represents income from a substantial business interest, which most often denotes a shareholding of at least 5% in a company. Box 2 income is subject to a flat tax of 25%. Finally, Box 3 combines income from savings and investments. The total amount in this Box can be allocated optimally between fiscal partners. Individually declared income in Box 3 over €19,522 is subject to a 30% flat tax, which is

taken on a fixed assumed return of 4% of the average yearly net value of the assets minus any liabilities. Because the tax is based on the average Box 3 value over the year, an increase of x in a given year from a baseline of y , the tax would be $0.3 \cdot 0.04 \cdot (y - 19,522) + 0.3 \cdot 0.04 \cdot x / 2$, which amounts to a 0.6% wealth tax on increases in Box 3 holdings.

Table 1: Income Tax in the Netherlands for 2005

Category	Bracket (€)	Tax Rate
Box 1: Income from home and work		
Profits from business or professional activities, main employment, other activities	SS: 0–30,357 if age <65	32.60%
Periodic payments (pensions, life annuity)	+	
Owner-occupied dwelling (imputed rent), mortgage interest	0–16,893	1.80%
Negative expenses for income provisions, deductions, etc.	16,893–30,357	9.35%
Deductions: commuting, childcare, other work-related expenses	30,357–51,762	42%
	51,762+	52%
Box 2: Income from substantial interest		
Dividends and capital gains from $\geq 5\%$ shareholding (direct/indirect, incl. partner) ¹	total share value	25%
Box 3: Income from savings and investments²		
Bank/savings accounts, stocks, second homes, rental income, etc.	$\max\{0, (\text{total}^3 - 19,522) \times 4\%\}$	30%
Deductions: debt interest, education, charitable contributions		

¹ If either taxpayer or fiscal partner holds a substantial business interest ($\geq 5\%$), all shares are taxed under Box 2, even if one partner holds less than 5%. If neither reaches 5%, neither is taxed in Box 2.

² Box 3 income may be allocated between partners, but tax is levied individually.

³ Box 3 total taken as the mean of beginning- and end-of-year Box 3 net value of the assets minus any liabilities.

Anyone liable to pay taxes for year t in the Netherlands is required to fill in their declarations by April 1st of year $t + 1$. If individuals do not submit a tax return, one is filled in automatically using available third-party information, including income, property, bank, and other financial records. Some components of third-party information, such as savings held in domestic banks, are reported regularly, while others, such as information on offshore bank accounts, may vary from year to year depending on international banking agreements. The tax authorities then begin analyzing tax declarations in July of year $t + 1$ and generate audit flags. Flags are based on characteristics of the returns, previous flags, and discrepancies between declarations and third-party information. It takes some time before the letters are sent out to the people whose tax declarations will be subject to audit. In most years, these letters are sent out between October of year $t + 1$ and September of year $t + 2$.² When an auditor detects misreporting, the taxpayer is required to pay the full outstanding sum. If the underreporting is deemed intentional cheating, an additional fine of between 50%-100% of the evaded amount may be imposed. For underreporting due to negligence, the fine is 25%.

2.2 Auditing Announcements

In addition to common auditing practices - targeted criteria, thresholds, random selection - the Netherlands Tax and Customs Administration selects a spotlight topic each year to be the subject of additional scrutiny. The motivation for choosing spotlight topics varies each year. In some years, the topic is selected based on suspicions of high levels of fraud, while in other years the intention is to educate the public on complex tax topics. The spotlight theme for fiscal year t , which follows the calendar year, is always announced to the public in the first days of

²For the 2007 topic, the sample of people due for audit was initially too big to handle. As a result, letters were sent out after mid-April 2009.

January of year $t + 1$.

Notifications of the spotlight topic are publicized through newspapers, magazines, radio, television, and internet announcements, and are made evident on the front page of the tax authorities' website.³ The spotlight announcement auditing threat is not cheap talk. The auditing campaign for the 2005 topic involved a thorough preliminary screening of all individuals liable, through third-party information or self-declaration, to pay taxes on income from side earnings and a full audit of 25% of these declarations. The auditing campaign in 2007 involved stricter preliminary screening than in other years and saw a hundred-fold increase in the number of full audits for topics in Box 3. Where misreporting is detected, taxpayers may face fines of up to 100% of the evaded amount. Appendix A documents the full set of spotlight topics from 2002 to 2011 and shows that most announcements concerned entirely different issues, unrelated to the themes of the 2005 and 2007 campaigns or the dividend tax cut. The practice of single-topic announcements was discontinued in 2011 and replaced in 2012 with more targeted, risk-based audits.

2.3 Announcements in 2005 and 2007

The main focus of this paper is the auditing announcement of 2005 and, to a lesser extent, the auditing announcement of 2007. The 2005 announcement targeted item 11 in the Dutch tax returns, *income from side earnings*, one of several items that fall under Box 1, the part of the tax return related to income. This category relates to secondary earnings and is separate from a person's main income, whether as an employee or an entrepreneur. As presented in the original tax forms and associated explanations in the Online Appendix, what falls under this category is broad. 'Side earnings' can range from 'performing as an artist or professional athlete' to 'providing one-off consultancy or advice'. It also includes ambiguously defined earnings from 'working for foreign clients' or 'managing assets in a way that involves more work than typical passive investment'. Related to this last point, additional remarks detail this management of assets can include: 'renovating and then selling parts of a property as separate units', 'personally carrying out significant maintenance or alterations on a rental property', and 'using inside information or specialist knowledge to generate income'.

The category, *income from side earnings*, is also linked to another item falling under Box 1 and other sections (Boxes) of the tax forms. If taxpayers fill in declarations for *income from side earnings* (item 11 of the Dutch tax forms), they are also directed to complete an additional entry (item 13) which elicits declarations from a second category in Box 1: *profits from assets made available* (item 12). *Profits from assets made available* include earnings from assets (e.g., a building) made available to companies, related persons⁴, or partnerships for business or other income-earning activities.

³For the 2005 announcement, in addition to the widespread campaign, letters were sent to the 181,551 taxpayers who declared *income from side earnings* in 2004. The letter informed people to take particular care in filling their returns for income from secondary sources such as side earnings income, which falls within Box 1. The Box 3 announcement in 2007 emphasized the categories of additional property, personal assets, savings and investments.

⁴In Dutch tax law, related persons include not only close family members and legal partners (e.g., tax partners or spouses when filing individually, registered cohabitants, minor children) but also individuals or entities with whom the taxpayer has qualifying personal or economic ties. This can extend to certain business relations — such as companies, partnerships, or other arrangements — in which the taxpayer or a related person holds a substantial interest or participates in a profit-sharing capacity.

The additional entry (item 13) asks taxpayers: “*Did you fill in item 11 [income from side earnings] or item 12 [profits from assets made available]?*” and “*In 2005, did you own other property linked to your freelance income, side earnings, or assets made available for use, or did you have other business assets or debts, each worth at least €2,500?*” If the answer is “yes” to both, taxpayers are required to report the beginning- and end-of-year values for three categories: *other property*, *other assets*, and *debts*. These same categories also appear independently in Box 3. The distinction between whether such items should be reported in Box 1 or Box 3 is ambiguous. The official explanatory document indicates that taxpayers retain some discretion: “*If, for example, you have used your own assets in your activities, you often have to **decide** whether to classify these assets as business or as private.*”⁵ If assets are deemed business-related, they must be declared in Box 1 (or in Box 2 if involving a substantial business interest, defined as a shareholding of over 5%). Otherwise, they are reported in Box 3.

The 2007 announcement targeted all items in Box 3.

A central claim of this paper is that the ambiguous classification of certain assets - such as private versus business use, or passive versus active management - provides taxpayers with discretion over how to allocate wealth across the tax form. As section 3 discusses, this discretion can be strategically exploited when the risk of audit increases for one part of the return but not others.

2.4 2007 Dividend Tax Cut

The 2007 dividend tax cut was designed as a short-term compensation for directors and major shareholders as an interim response to new legislation. In 2006, the Dutch government introduced a Health Insurance Act, which required employees to pay a 6.5% contribution on their wages while self-declared entrepreneurs paid a lower rate of 4.4% on their taxable business profits. This discrepancy led to a higher burden on large shareholding company directors, who fall under the category of ‘employees’, rather than ‘entrepreneurs’. This difference was deemed unfair but could not be rectified before 2008 due to technical difficulties. In response, the government temporarily reduced the Box 2 tax rate by 3 percentage points in 2007, lowering the tax on returns from substantial interest from 25% to 22% for declarations up to €250,000.

3 Conceptual Framework

We consider three types of taxpayers: (i) avoidance, (ii) non-optimizing, and (iii) evasion. Here, ‘avoidance’ refers to legal tax-minimizing behavior, while ‘evasion’ refers to the concealment of income or assets. ‘Non-Optimizing’ means that the taxpayer is not responding rationally to prevailing tax rates and audit probabilities. We also assume that the avoidance group already minimize their tax burden have no reason to adjust their filings in response to the 2005 spotlight announcement.

The remaining two groups—non-optimizing honest taxpayers and evading taxpayers—are expected to react differently. For non-optimizing honest taxpayers, the 2005 announcement may serve as an informational shock. Due to the salience (Chetty et al., 2009) of the announcement, taxpayers who had previously declared ambiguous assets in Box 1 may revisit the legal boundary between Box 1 and Box 3 (or Box 2) declarations and learn that it is fiscally advantageous to

⁵Translation from the official explanatory documents for the Dutch tax return, 2005. English rendering via ChatGPT, with manual corrections, added emphasis, and explanatory bracketed text by the author.

report these assets elsewhere. This group may therefore reduce declarations in Box 1 categories such as *income from side earnings* and *profits from assets made available*, while increasing declarations in Boxes 2 and 3. These changes are expected to persist over time, reflecting learning effects rather than a purely transitory response. That said, this channel may be limited in scope, particularly among taxpayers who were already using tax advisers prior to 2005.

The incentives for evading taxpayers are more involved. These taxpayers may correctly infer that the probability of being audited in 2005 increases with the level of declarations in *income from side earnings* and the associated category *profits from assets made available*. They may also anticipate that audit resources will be concentrated on high-income or high-wealth individuals. Consequently, if they choose not to disclose previously hidden wealth in 2005, they risk detection and retroactive penalties. In response, these taxpayers may opt to disclose evaded income or assets, but strategically reallocate them to Box 3 (or Box 2) rather than Box 1.

This reallocation serves several purposes. Most importantly, above a relatively low monetary threshold, it minimizes tax burden.⁶ It also allows taxpayers to declare income without triggering the audit algorithms associated with the 2005 spotlight category. Finally, by avoiding a spike in Box 1 entries, taxpayers reduce the risk of retrospective scrutiny, even if the 2005 declaration itself is honest. In contrast to the learning-driven behaviour of non-optimising taxpayers, evaders are not expected to reduce Box 1 declarations. Rather, their response will be an increase in Box 3 (and possibly Box 2) declarations, representing the disclosure of what would otherwise be hidden wealth. [López-Luzuriaga \(2021\)](#) discuss similar strategic behavior to avoid scrutiny in the context of tax authorities forcing taxpayers to itemize deductions.

A priori, it is not clear which of these two groups, avoiders and evaders is larger. If one assumes most taxpayers influenced by the 2005 are high earning tax-payers, as we show is the case in our setting, then we might assume the avoidance group is relatively small. This is because most of these taxpayers use tax advisers who would, presumably, already minimize their tax burden.

In 2007, incentives differ. Honest taxpayers are not expected to change their reporting in response to the Box 3 spotlight announcement. Evading taxpayers may no longer see the benefit in reallocating declarations from Box 3 to Box 1, as they could in 2005. Not only would this move incur a significantly higher tax burden, but the audit threat may appear less salient. Since roughly 90% of taxpayers report Box 3 income, the probability of being singled out may be perceived as lower than under the more targeted 2005 campaign.

As a result, the 2007 announcement may lead to small increases in Box 3 declarations, particularly for topics with poor third party reporting such as *other property*, *other assets*, and *debts*. At the same time, both honest and evading taxpayers may respond to the one-year dividend tax cut in 2007. Those with substantial shareholdings (above 5%) may report higher dividends in Box 2, while others may reflect these changes in Box 3 categories such as *shares and bonds* or *savings*.

⁶Box 3 does not minimize taxes for all taxpayers because Box 1 can yield a lower liability for low-income individuals or for low-return, cost-intensive assets. For high-income taxpayers facing marginal Box 1 rates of 42 to 52 percent, however, any asset with a moderate return exceeds the point at which Box 3's effective 1.2 percent wealth tax becomes cheaper than taxing actual income in Box 1. Since the taxpayers in our sample overwhelmingly fall into these higher brackets, the observed shift toward Box 3 is consistent with tax-minimizing behavior in this population.

4 Data

The analysis uses longitudinal data from the Netherlands Tax and Customs Administration for 2002-2008, which include annual individual tax declarations as well as individual socio-demographic and employment characteristics.⁷

We consider three samples for the analysis. The first is a representative sample of 65,005 individuals, followed from 2002 to 2008, who filed their taxes. The other two samples are focused random samples of individuals whose incentives are most likely to be affected by the 2005 auditing announcement. More precisely, we draw a random sample of 33,666 individuals from the pool of taxpayers declaring either positive or negative returns in the Box 1 category of *income from side earnings*. We further draw a random sample of 50,147 individuals from those declaring the linked Box 1 category of *profits from assets made available*. Appendix B provides tables describing and comparing these samples.

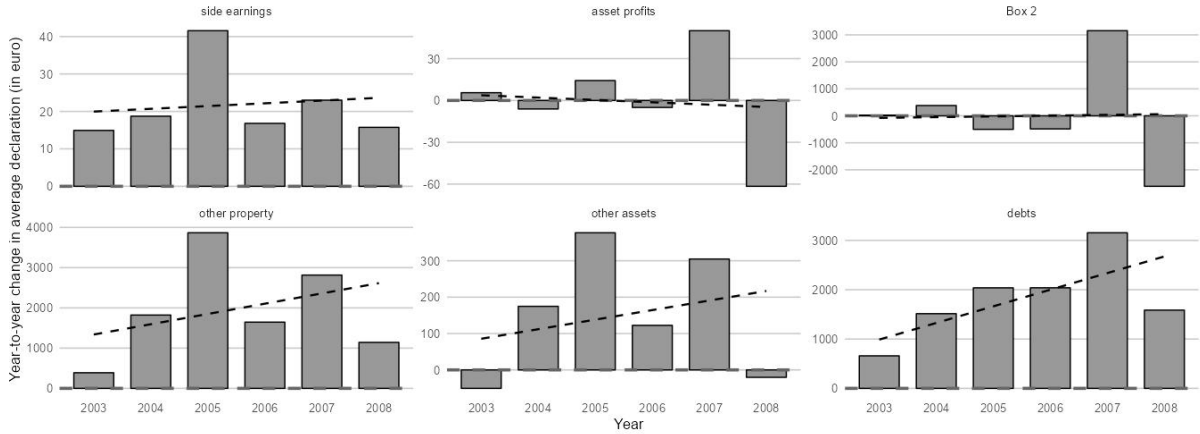


Figure 1: Yearly changes in average declarations, 2002–2008

Notes: This figure is constructed from Dutch administrative longitudinal tax records covering a representative sample of 65,005 individuals who filed taxes in each year from 2002 to 2008. The plotted values show annual averages of the first difference in declared amounts for each asset category presented in the paper. The dashed line overlays the baseline linear trend estimated in our main specification, illustrating the underlying model-implied trajectory against observed changes.

Figure 1 presents the average yearly change in declarations for our main outcomes of interest in the representative sample of taxpayers.⁸ We observe distinctly higher changes in average declarations of *income from side earnings*, *other property*, and *other assets* in 2005, and increases for the latter two categories in 2007. We also observe a large jump in average yearly declarations of *Box 2* and *profits from assets made available* in 2007, followed by proportional drops in 2008.⁹ We further present the plots of these variables in levels in Figure A2 of Appendix B.1.

Figure 2 considers in greater detail the strategic response to the 2007 tax rate cut for the first €250,000 of Box 2 declarations. It compares the distribution of Box 2 declarations

⁷The Dutch tax system was overhauled in 2001, so that year is excluded as an unusual learning year of the new tax code. Furthermore, the financial crisis hit the Netherlands most seriously in 2009, which severely affected trends in categories related to wealth.

⁸The categories that can be negative are Box 1, Box 2, IFW, PAA, and savings. If the subtopics in Box 3 sum up to a negative amount, the overall declaration is set to 0.

⁹Figure A1 of Appendix B.1 further presents aggregate trends for other relevant tax categories - *shares-bonds*, *savings*, *2nd home* - which may also be affected in a secondary manner by the 2005 and 2007 spotlight announcements and the 2007 dividend tax cut.

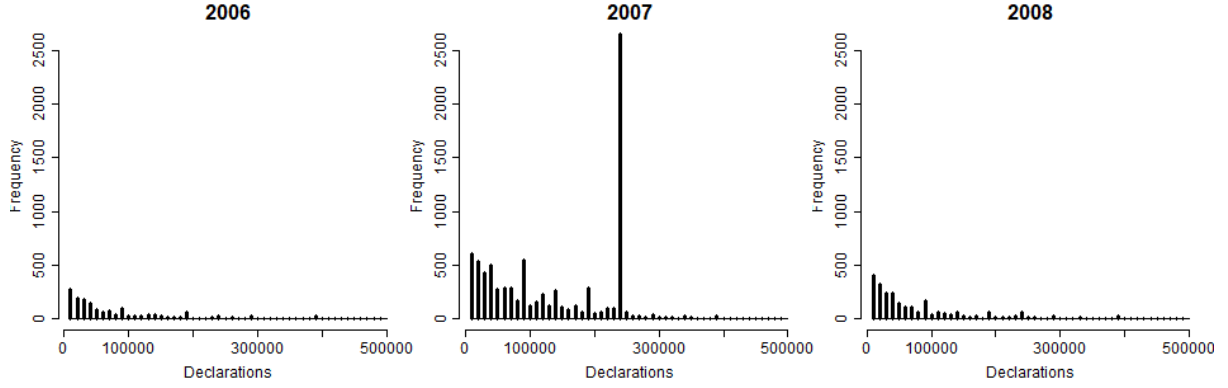


Figure 2: Box 2 declarations, 2006–2008

Notes: This figure plots the distribution of *positive* Box 2 declarations in 2006, 2007, and 2008 for individuals in the representative balanced sample of 65,005 taxpayers. Values are displayed only above €10,000 for presentational clarity.

above €10,000 in 2006, 2007, and 2008. The reaction to the dividend tax cut is evident from the bunching around the €250,000 threshold.¹⁰ In total, 2.75% of the representative sample declared positive Box 2 returns in 2007 as compared to 0.94% in 2006 and 1.24% in 2008.

5 Model and Identification

Because all taxpayers are exposed to the spotlight announcements and the dividend tax cut, we cannot formulate separate treatment and control groups. Instead, we take an interrupted time series approach, with the average yearly changes in declarations presented in Figure 1 informing our empirical specification. We model baseline declarations using all years from 2002–2008 other than 2005 and 2007, and estimate the effects of the announcements in 2005 and 2007 as year-specific bumps in declarations. More precisely, for taxpayer $i = 1, \dots, N$ in year $t = 1, \dots, T$, we model tax declarations on a given topic y_{it} using the following specification, which we estimate in first differences:

$$y_{it} = \alpha_i + f(t) + \beta_{2005} \cdot \mathbf{1}(\text{year} = 2005) + \beta_{2007} \cdot \mathbf{1}(\text{year} = 2007) + u_{it} \quad (1)$$

y_{it} denotes one of the outcomes that may be linked to the 2005 and 2007 announcements and the 2007 dividend tax cut, or outcomes that serve as placebo evaluations. These include *income from side earnings* and *profits from assets made available* in Box 1, Box 2 declarations, and every listed item falling under Box 3. After the first difference, we will have $\Delta \mathbf{1}(\text{year} = 2005) = 1$ and $\Delta \mathbf{1}(\text{year} = 2006) = -1$. Similarly, $\Delta \mathbf{1}(\text{year} = 2007) = 1$ and $\Delta \mathbf{1}(\text{year} = 2008) = -1$. We are interested in the parameters β_{2005} and β_{2007} , which approximate the average change in tax declarations in 2005 and 2007 relative to a baseline individual level plus a topic-specific trend given by $\alpha_i + f(t)$. α_i is an individual-specific intercept, while $f(t)$ is specified such that there is a linear time trend after taking the first difference. We also offer results using a third degree polynomial, a time trend estimated only on years before 2005, and a within estimator in

¹⁰88.2% of taxpayers with Box 2 declarations in 2007 between €240,000–€260,000 declared €0 Box 2 returns in 2006 and 79.1% of them declared €0 Box 2 income in 2008. These numbers are, however, not dissimilar to other taxpayers' changes in Box 2 declarations.

Appendix B.1.¹¹ We do not include time-varying demographic or employment covariates, since these may be bad controls endogenous to evasion.¹² The rationale for a common first-difference specification across outcomes is to ensure comparability and to avoid selectively fitting outcome-specific time trends. While some responses, particularly in Box 3, may plausibly persist beyond the announcement year, the first-difference specification places greater weight on short-run responses consistent with evasion incentives and therefore yields conservative estimates when effects are longer-lasting. In a setting with highly persistent declarations, the first-difference estimator is also more efficient than the within estimator.

The underlying assumption is that the baseline trend adequately captures the expected counterfactual declarations for 2005 and 2007. Speaking to this assumption, anticipation effects with respect to spotlight topics are unlikely since these are only announced internally one to two months prior to their public release. Announcements in other years also concerned tangential topics unlikely to affect those under study.¹³ With that said, the dividend tax cut applied to Box 2 was known in advance, allowing major company shareholders to decide on the optimal distribution of dividends ahead of time.

Additionally, this model ignores some types of long-run announcement effects in post-announcement years. Rectified under-reporting due to negligence would lead to underestimating the magnitude of β_{2005} and β_{2007} . These parameters would also be underestimated if the threat effect of the announcements were persistent beyond the initial announcement year, as shown for audit effects in [Boning et al. \(2023\)](#). Other post-announcement-year effects are integral to our study. These include ‘crater’ effects, whereby individuals sharply increase their evaded taxes after announcement years. A final problem is that announcement effects will still underestimate misreporting since we cannot capture undeclared funds from taxpayers who never react to announcements and keep returns hidden in all years.

6 Results

6.1 2005 Effects on Representative Sample

Figure 3 presents the main results for the representative sample of taxpayers, showing point estimates and 95% confidence intervals of β_{2005} and β_{2007} for each listed tax topic.

Despite the aggregate increase in declarations of the directly targeted *income from side earnings* in 2005 (see Figure 1), our main specification finds no statistically significant change in this category in the representative sample ($\hat{\beta}_{2005} = 12.742$ (9.142)). This result likely reflects the low prevalence of declarations: only around 4–5% of taxpayers report any income in this category in a given year, which limits the statistical power of estimates. We revisit this issue in Section 6.2, where we focus on a targeted sample of taxpayers who declared *income from side earnings* at least once during the 2002–2008 period. For now, we turn to the broader effects visible in other categories of declared wealth.

¹¹The third-degree polynomial runs the risk of overfitting the trend since, due to the low number of observed years, it perfectly interpolates the data.

¹²As is well-known, effects in levels can be pulled by relatively few large changes in declarations at the top end of the wealth distribution. As such, we also show results with a log specification of the outcome in Appendix B.3. In Appendix B.3, we also present the share of positive, zero (and negative) declarations for each outcome of interest which offers insights into changes at the extensive margin.

¹³Listed and discussed in Appendix 2.1.

Beyond the direct category, Figure 3 shows a clear increase in *other property*, one of the Box 3 items linked to the 2005 Box 1 spotlight topic. We also observe smaller increases in *other assets*, albeit with weaker statistical significance.¹⁴ These shifts in property-related wealth are consistent with recent work on wealth hidden in foreign real estate (Alstadsæter et al., 2022b; Bomare and Herry, 2022; Johannesen et al., 2022).

Notably, we observe no spillover into Box 2. This may reflect both financial disincentives, since Box 2 declarations are taxed at a higher rate than Box 3, and clearer legal boundaries, since Box 2 only covers returns from companies in which the taxpayer holds more than 5% of shares.

Extrapolating to the full population, the 2005 increase in *other property* declarations amounts to approximately €800 million. This should be interpreted as a shift in reported assets conditional on the validity of the interrupted time series specification. At prevailing tax rates, a back-of-the-envelope calculation suggests that this increase resulted in additional tax revenue of under €5 million. This number can be seen as an upper bound as it ignores the additional auditing and advertising costs of the spotlight announcement. If the same underlying assets had instead been taxed in Box 1, the implied annual tax on their income flows would be substantially larger. For instance, under a conservative 4% net return, as assumed in Box 3, and a 50% marginal Box-1 rate, the additional tax would be about €16 million.

Extrapolating to the full population, the 2005 increase in other property declarations amounts to approximately €800 million, implying additional Box-3 tax revenue of just under €5 million. If the same underlying assets had instead been taxed in Box 1, the implied annual tax on their income flows would be substantially larger. For instance, under a conservative 4% net return, as assumed in Box 3, and a 50% marginal Box-1 rate, the additional tax would be about €16 million.

Turning to other items not directly related to Box 1, we also see an increase in *savings* and, with lower statistical significance, in the *second home* category. However, as shown in Appendix B.3, Figures A4 and A3, these results are less robust when using log specifications or higher-order polynomial trends.

Appendix B.3 Figures A5 to A6 further presents additional specification results using a within estimator and when estimating the underlying trend only on pre-2005 years. The results remain robust.

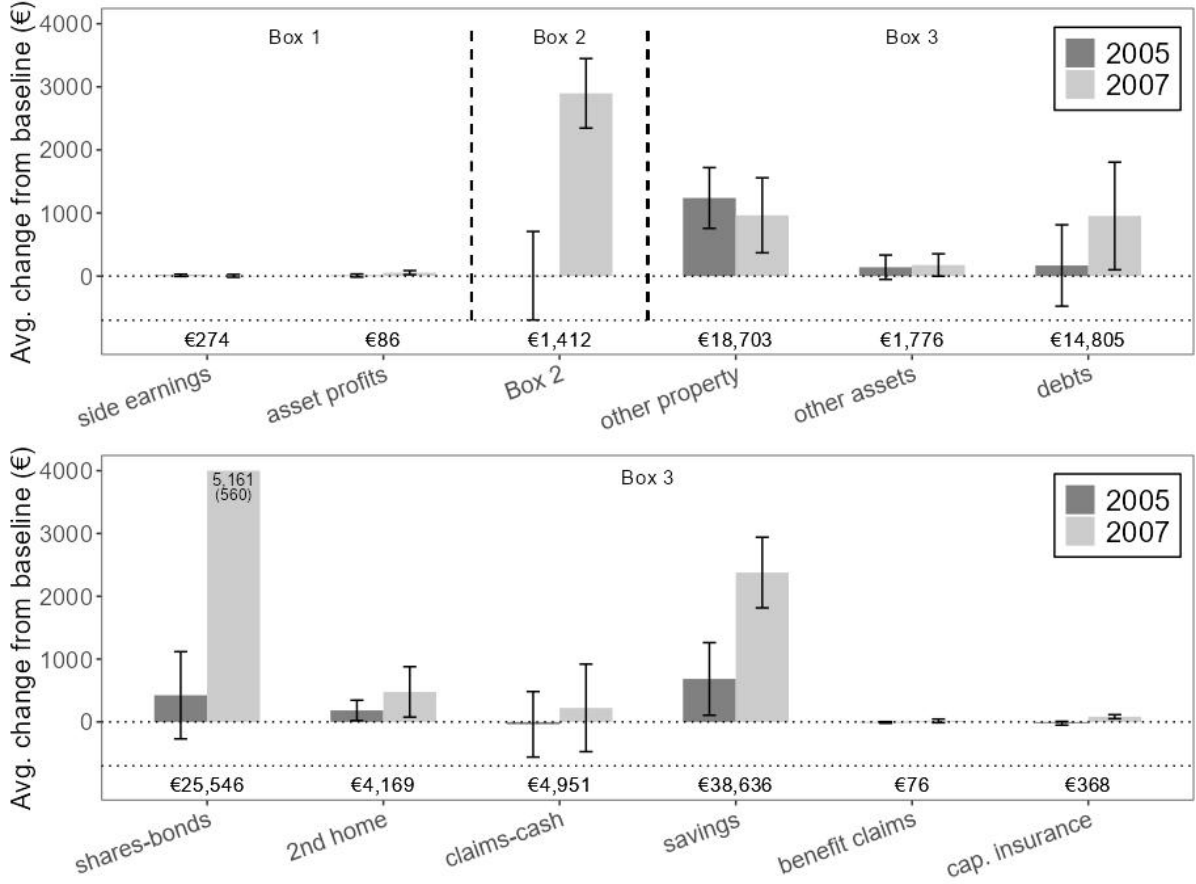
6.2 2005 Effects in the Side Earnings Sample

We now focus on a sample restricted to taxpayers who declared any *income from side earnings* during the 2002–2008 period. In this targeted sample, Figure 4 shows a visible increase in declarations in 2005, consistent with the pattern in Figure 1. Table 2 confirms this effect in modeled estimates, with an average increase of about €100 in declared income, corresponding to roughly 4% of the baseline mean¹⁵ which is also reflected in the log specification results.

¹⁴We see no significant effect on *debts*, the third item that may be declared under all Boxes. This may reflect counteracting forces. Taxpayers might over-declare deductible debts in normal years, but reduce them when the probability of audit rises. At the same time, debt declarations often accompany declarations in *other property* and *other assets*, for example through mortgage liabilities, which would mechanically increase as those categories increase. See Klepper and Nagin (1989) for similar patterns and explanations in U.S. TCMP data.

¹⁵The baseline mean excluding the values of 2005 and 2007 is €2770.

Figure 3: Representative sample declaration effects in 2005 and 2007

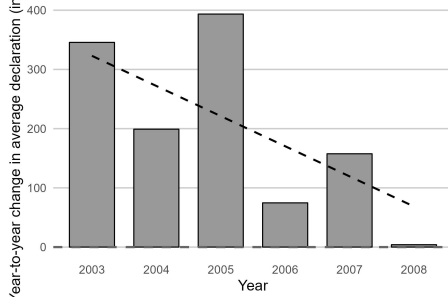


Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level regressions of first differences in declared amounts on year indicators, as specified in Equation 1. Units are euros per taxpayer. Bars show the estimated change in average declarations relative to the linear time trend and individual fixed effects. For Box 2 in 2007, the bar is truncated for presentation purposes; the exact estimate is €2,897 with a standard error of €695, and these values are printed on the truncated bar in the figure. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each outcome over 2002–2008, excluding 2005 and 2007, for the balanced panel of 65,005 individuals.

When extrapolated, this implies an aggregate increase of approximately €3 million.

¹⁵To assess the additional effect of letters sent to taxpayers who declared *income from side earnings* in 2004, Table A5 in Appendix B.2 estimates the change in declarations at t conditional on having declared non-null income at $t - 1$. The results show that *income from side earnings* is typically transitory. Taxpayers who declare it in $t - 1$ are less likely to do so in t .

Figure 4: Yearly changes in side earnings declarations, 2002–2008, *income from side earnings* sample



Notes: The figure reports annual average year-on-year changes in declared income from side earnings (Box 1) for random sample of taxpayers who declared positive side earnings at least once during 2002–2008 (33,666 individuals). The dashed line overlays the baseline linear trend estimated in our main specification, illustrating the underlying model-implied trajectory against observed changes.

Table 2: Estimated effects of the 2005 announcement on side earnings declarations

	Level	Log	Level, 3rd-deg. trend
Effect	100.012* (51.331) [0.051]	0.041*** (0.014) [0.003]	182.312* (103.957) [0.079]
Mean	2,895.608	2,895.608	2,895.608
Obs.	235,662	235,662	235,662
Ind.	33,666	33,666	33,666

Income from side earnings sample (taxpayers with positive side earnings at least once during 2002–2008). Entries report coefficients for the 2005 indicator in first-difference specifications of Equation 1, with a linear time trend or a third-degree polynomial trend. Outcomes are declared income from side earnings, in euros per taxpayer (Level columns) or in logs. Standard errors, in parentheses, are clustered at the individual level; p-values are in brackets. *p<0.10, **p<0.05, ***p<0.01.

6.3 Interpretation and Behavioural Implications

Taken together, these findings support the behavioural mechanisms laid out in Section 3. The observed increases in Box 3 declarations, particularly for ambiguous categories such as *other property*, are consistent with dishonest taxpayers strategically reallocating wealth across boxes to reduce audit risk and tax liability, as well as with non-optimising honest taxpayers updating their understanding of optimal reporting.

This behaviour differs from what is observed under tax amnesties. For example, [Alstadsæter et al. \(2022a\)](#) find no strong substitution between evasion and avoidance in response to an amnesty in Norway. Two features of our setting may explain the difference. First, unlike amnesties, disclosure in response to heightened audit risk can expose taxpayers to potential retrospective penalties for prior-year evasion. Second, the Dutch tax code allows certain assets and returns to be declared under different tax bases depending on factual characterisation (e.g. business-connected versus private use), so increased scrutiny of a specific category can induce reclassification without changing the underlying asset. Norway offers fewer comparable within-return classification margins: assets and returns are more mechanically assigned to capital income and wealth tax bases, and avoidance operates primarily through ownership structure or timing rather than reclassification.

6.4 2007 Effects on Representative Sample

The changes in 2007 combine the effects of two policy events: the Box 3 spotlight announcement and a dividend tax cut affecting Box 2. The most pronounced result is a direct increase in Box 2 declarations, with an average jump of approximately €2,897 per taxpayer, an effect already

documented in Bettendorf et al. (2017).¹⁶

Because dividend payouts also reach shareholders holding less than 5% of shares, we observe broader shifts across the tax base. Most notably, *profits from assets made available* in Box 1 increase by an average of €55. This likely reflects fiscal incentives for directors–large shareholders to report limited profits in Box 1, as discussed in Bettendorf et al. (2017).

We also detect shifts in the Box 3 categories of *other property*, *shares and bonds*, *savings*, and *debts*. However, only the increases in *savings* and *debts* are robust to alternative trend specifications, including log outcomes and third-degree polynomials (Appendix B.3, Figures A4 and A3). The increase in *savings*, a category largely subject to third-party reporting, is unlikely to be driven by the Box 3 spotlight announcement. It more plausibly reflects spillovers from widespread dividend payouts to minor shareholders.

In total, the dividend tax cut corresponds to a €1.9 billion increase in Box 2 declarations in 2007. Accounting for increases in *savings* and offsetting the rise in *debts*, the estimated total increase in declarations amounts to around €2.8 billion.¹⁷ A back-of-the-envelope calculation suggests these changes combined to increase tax revenue by €480 million in 2007.

While the sharp increase in Box 2 revenue was not anticipated by policymakers, it did not go unnoticed.¹⁸ In 2014, facing a balance of payments deficit, the Dutch government reduced the Box 2 tax rate to encourage further dividend distributions. This produced the expected effect of raising tax revenue from Box 2 dividend payouts. A similar spike occurred in 2019, this time in anticipation of a planned increase in the Box 2 tax rate from 25% to 26.9% and a proposed tightening of rules on borrowing from one’s own company.¹⁹

6.5 Temporary or Lasting Changes in Declarations?

Table 3 examines whether the behavioural responses to the 2005 audit announcement and the 2007 dividend tax cut were temporary or persistent. To do so, we modify the main specification by including separate indicators for the announcement and the post-announcement year, and estimating the 2005 and 2007 effects separately on relevant outcomes.

More specifically, for the categories affected by the 2005 announcement we estimate the first difference model:

$$\Delta y_{it} = \Delta f(t) + \beta_{2005} \cdot \mathbf{1}(year = 2005) + \beta_{2006} \cdot \mathbf{1}(year = 2006) + \Delta u_{it} \quad (2)$$

This specification is equivalent to a levels model with step changes in 2005 and 2006. β_{2005} captures the initial jump in declarations, while β_{2006} captures whether that change persists, dissipates, or reverses in the following year.

For 2005, the spike in declarations of *other property* and *other assets* does not appear to reverse in 2006. The estimated effects for 2006 are near zero and not statistically significant, which may be a sign of some learning effects.

In contrast, the 2007 effects, particularly in Box 2, show clear signs of a temporary increase

¹⁶Bettendorf et al. (2017) focus on shifting behaviour among directors–large shareholders (DGAs) who hold more than 5% of shares in a firm.

¹⁷The implied elasticity of Box 2 declarations is approximately 0.6.

¹⁸While not the focus of this study, the repeated jumps in Box 2 declarations are relevant to studies focused on profit retention by directors–large shareholders discussed, among others, in (Chetty and Saez, 2006; Yagan, 2015; Alstadsæter et al., 2017; Schjelderup and Zoutman, 2024).

¹⁹These changes were ultimately implemented in 2023.

Table 3: Temporary or lasting changes in declarations?

	2005-2006 reversal					2007-2008 reversal		
	Side earn.	Asset prof.	Other prop.	Other assets	Debts		Asset prof.	Box 2
2005 effect	23.709*	14.824	2424.048***	282.782**	450.276	2007 effect	50.687*	3899.383***
	(12.022)	(19.593)	(419.396)	(121.264)	(458.511)		(27.636)	(695.374)
	[0.049]	[0.449]	[0.000]	[0.020]	[0.326]		[0.067]	[0.000]
2006 effect	-1.540	0.463	4.223	12.417	169.928	2008 effect	-59.760**	-1627.039***
	(13.000)	(16.430)	(361.378)	(135.833)	(515.379)		(27.042)	(529.500)
	[0.906]	[0.978]	[0.991]	[0.927]	[0.742]		[0.027]	[0.002]
Mean	286	97	19820	1888	15603		286	1,777
Obs.	455,035	455,035	455,035	455,035	455,035		455,035	455,035
Ind.	65,005	65,005	65,005	65,005	65,005		65,005	65,005

Notes: The table reports coefficients from separate first-difference regressions estimating discrete year effects using Equation 2. Coefficients are in euros per taxpayer. Standard errors (in parentheses) are robust and clustered at the individual level; p-values are reported in brackets. *p<0.10; **p<0.05; ***p<0.01. The left panel compares 2005 and 2006 to the baseline trend to assess whether changes associated with the 2005 spotlight announcement persist or reverse. The right panel analogously estimates 2007 and 2008 effects following the dividend tax cut and Box 3 spotlight announcement.

in declarations when using an analogous specification for 2007. The large positive jump in 2007 is followed by a significant negative coefficient in 2008 (€1,627), consistent with higher dividend payouts to take advantage of the temporary rate cut. Our analysis cannot, however, confirm whether this increase represents intertemporal shifting or simply a one-year increase in dividend payouts. A similar, albeit smaller, reversal is observed in asset profits in Box 1.

6.6 Heterogenous Effects

Table 4 investigates heterogeneity in behavioural responses to the 2005 audit announcement and the 2007 dividend tax cut, focusing on selected outcomes for the representative sample. Specifically, we examine changes in *other property* (Box 3) and *profits from assets made available* (Box 1) in 2005, and in Box 2 declarations in 2007.

The results show that the 2005 spike in *other property* declarations occurred among both lower- and higher-income taxpayers, though the increases were markedly larger for those with declared earnings above €100,000. In contrast, the 2007 increase in Box 2 declarations, and in *profits from assets made available* in Box 1, is driven almost exclusively by the high-income group. Wealthier individuals also exhibited greater increases in Box 3 *other property* declarations in 2007. This could reflect either spillovers from dividend payouts to non-majority shareholders - misclassified as *other property* or *profits from assets made available* - or a direct response to the Box 3 audit announcement. As previously discussed, unlike in 2005, where Box 3 offered a low-tax outlet for previously hidden assets, the only remaining alternative in 2007 was Box 1, which carries much higher marginal tax rates. Rational evaders may therefore have chosen to declare assets transparently in Box 3 despite the spotlight.

Columns 4–6 explore heterogeneity by tax professional usage. Although wealthier taxpayers are more likely to use tax professionals (68% versus 46% for those below €100,000), not all high earners do so, and vice versa. Taxpayers listing a tax professional reported increases in *other property* in 2005 that were roughly twice as large as those of taxpayers without a professional, though both groups experienced positive and significant changes. This gap may stem from

supply-side frictions: some advisors may discourage reporting previously concealed wealth.²⁰

We also examine gender differences. Box 2 declarations in 2007 rose for both men and women, but increases were significantly larger for men. In 2005 and 2007, only men significantly increased declarations in *other property* and *profits from assets made available*, respectively. These differences likely reflect gender disparities in wealth: mean earnings for men in the sample are €43,348, more than double that of women (€18,358).

Finally, we explore heterogeneity by entrepreneur status.²¹ The 2007 increase in Box 2 is concentrated among non-entrepreneurs, consistent with the policy’s design to compensate non-entrepreneurial shareholders for inequities in the 2006 Health Insurance Act. We find no evidence that taxpayers switched out of entrepreneur status to benefit from the dividend tax cut.²² In 2005, *other property* responses do not significantly differ between entrepreneurs and non-entrepreneurs.

Overall, these heterogeneity results reinforce the interpretation that audit and policy responses are concentrated among wealthier individuals. To further investigate this, the next section isolates a specific group of taxpayers whose incentives are particularly likely to be affected by the audit announcements and the 2007 dividend tax cut.

6.7 2005 and 2007 Effects on Box 1 Other Profits Sample

The previous results present shifts and reactions for a representative subset of taxpayers in the population. Figure 5 considers a sample particularly likely to react to the 2005 announcement. It focuses on a random sample of taxpayers who declared *profits from assets made available* in Box 1 in 2002-2008, the category linked to the targeted topic of the 2005 announcement. These individuals are especially likely to have been aware of and responsive to the announcement. The results, shown in Figure 5, reveal pronounced changes in declaration behaviour in both 2005 and 2007 for this group.

The patterns mirror those observed in the representative sample but are substantially larger in magnitude. This is not surprising given the characteristics of the sample presented in Appendix B. Notably, 33.6% (10.9%) of individuals in this sample report earnings, defined as the sum of employment income and annual returns from wealth, above €60,000 (€120,000), compared to just 10.1% (2.2%) in the representative sample. Furthermore, 45.7% of them work in the financial sector (versus 13.5%), and 83% use tax professionals (versus 39.8%). They are also much more likely to report in other Box 2 and Box 3 categories: 28.2% declare *other property* (versus 9%), 4.9% declare *other assets* (versus 2%), and 47.2% declare *debts* (versus 19%). These patterns reinforce the interpretation that compliance responses were concentrated among high-wealth, professionally advised taxpayers.

7 Policy Insights

While a complete welfare or revenue impact analysis is beyond the scope of this paper due to the unavailability of crucial administrative data, such as audit costs and recovery rates for different audit types, it is nevertheless possible to draw several policy-relevant insights from

²⁰Appendix Table A6 shows no substantial increase in tax professional usage in either 2005 or 2007.

²¹Under Dutch tax law, individuals are classified as ‘entrepreneurs’ for income tax purposes if they independently operate a business and bear the associated risks. Director-shareholders typically do not qualify.

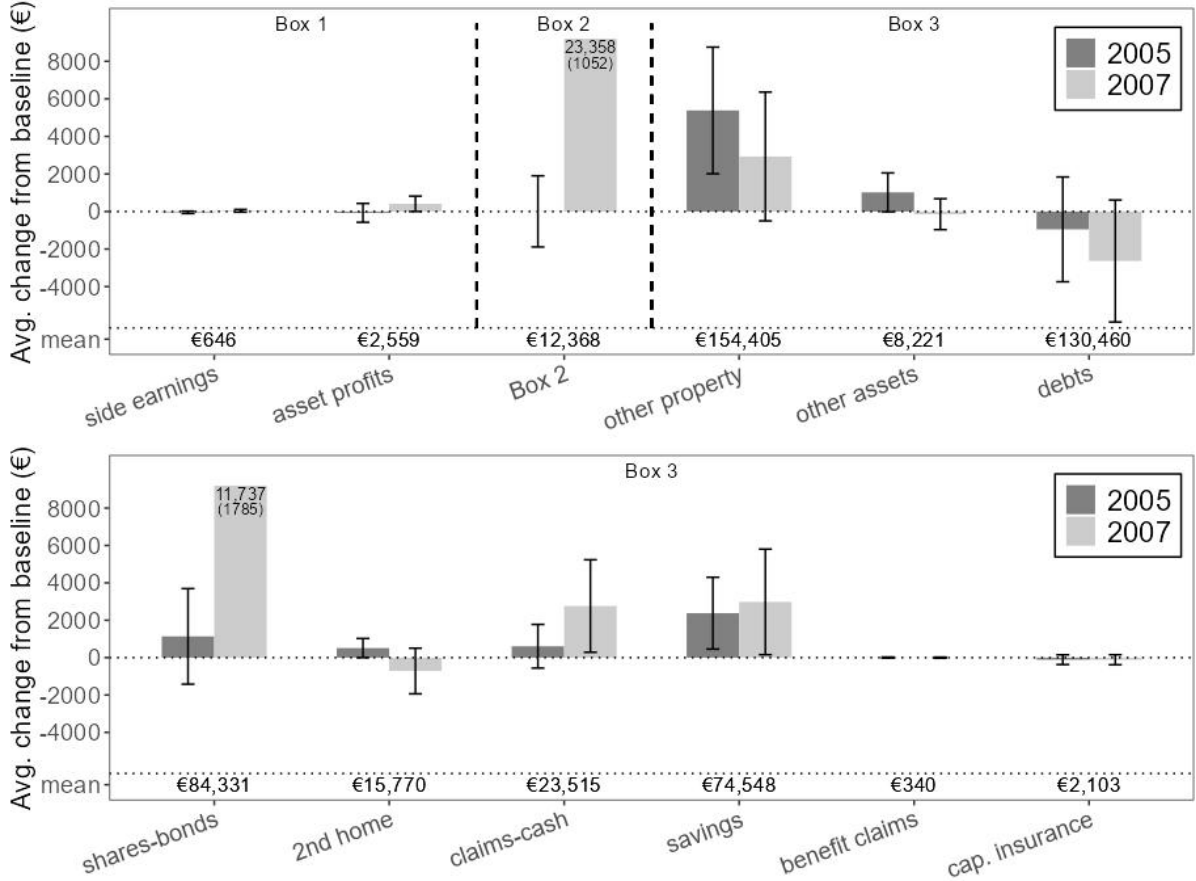
²²Appendix Table A6 shows no change in the share of taxpayers reporting as entrepreneurs.

Table 4: Heterogeneous Effects on Tax Declarations

	Wealth above €100k			Tax professional			Entrepreneur			Male		
	Box 2	other property	asset profits	Box 2	other property	asset profits	Box 2	other property	asset profits	Box 2	other property	asset profits
2005 effect	66.367 (75.183) [0.377]	924.181*** (199.643) [0.000]	10.130 (8.938) [0.257]	23.674 (62.705) [0.706]	759.075*** (157.666) [0.000]	-5.635 (4.046) [0.164]	-55.056 (403.841) [0.892]	1252.286*** (249.789) [0.000]	8.885 (13.414) [0.508]	-21.022 (74.596) [0.778]	254.322 (268.164) [0.343]	7.490 (10.493) [0.475]
2007 effect	-96.600* (54.823) [0.078]	500.732* (276.505) [0.070]	7.603 (10.902) [0.486]	486.365*** (87.344) [0.000]	493.563*** (132.601) [0.000]	3.718 (3.896) [0.340]	3185.829*** (320.326) [0.000]	828.872*** (313.379) [0.008]	49.278*** (16.879) [0.004]	1648.435*** (124.257) [0.000]	406.409 (461.774) [0.379]	8.882 (8.753) [0.310]
2005 het. effect	-1270.143 (7027.159) [0.857]	6596.107*** (3100.574) [0.033]	-27.992 (181.397) [0.877]	-37.389 (714.603) [0.958]	1005.794** (488.622) [0.040]	29.033 (24.709) [0.240]	498.096 (447.432) [0.266]	-114.681 (912.290) [0.900]	-0.720 (33.120) [0.983]	54.655 (692.338) [0.937]	2000.405*** (472.905) [0.000]	2.654 (23.847) [0.911]
2007 het. effect	62856.018*** (6038.796) [0.000]	9723.292*** (3346.357) [0.004]	994.063*** (265.935) [0.000]	5059.948*** (618.052) [0.000]	987.063 (660.228) [0.135]	107.530*** (34.354) [0.002]	-2361.266*** (414.113) [0.000]	1103.409 (1082.136) [0.308]	46.365 (58.716) [0.430]	2538.508*** (599.146) [0.000]	1133.258* (629.880) [0.072]	93.655*** (33.240) [0.005]
Mean	12,368	154,405	2,559	12,368	154,405	2,559	12,368	154,405	2,559	12,368	154,405	2,559

Note: Table displays estimated coefficients from first-difference regressions on indicators for 2005 and 2007 and their interactions with the listed indicator variables, using Equation 1 as the baseline specification. The rows labelled ‘2005 effect’ and ‘2007 effect’ report the effects for the baseline group (indicator = 0), while ‘2005 het. effect’ and ‘2007 het. effect’ report the interaction effects with the corresponding indicator: Wealth above €100k, Tax professional (used a tax professional to file), Entrepreneur, and Male. For individuals with indicator = 1, the total effect is given by the sum of the baseline effect and the corresponding heterogeneous effect. Coefficients are in euros per taxpayer. Standard errors, in parentheses, are robust and clustered at the individual level; p-values are reported in brackets. Number of observations: 455,035. Number of individuals: 65,005. * p<0.10; ** p<0.05; *** p<0.01.

Figure 5: Other Profits sample declaration effects in 2005 and 2007



Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level regressions of first differences in declared amounts on year indicators, as specified in Equation 1. Units are euros per taxpayer. Bars show the estimated change in average declarations relative to the linear time trend and individual fixed effects. For the *Box 2* and *shares-bonds* estimates of β_{2007} , the bars are truncated for presentation purposes; exact estimates and standard errors are printed directly on the truncated bars in the figure. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each outcome over 2002–2008, excluding 2005 and 2007, for the *other profits* sample (taxpayers declaring positive other profits at least once during the period).

the empirical findings. This section outlines three core implications based on the behavioural responses identified in the analysis.

Bundling Ambiguous Assets into a Unified Tax Category

The results show that ambiguity in asset classification, particularly across Boxes 1, 2, and 3, facilitates strategic declaration behaviour. Taxpayers appear to exploit overlaps in asset definitions to minimise liability, reallocating wealth in response to audit threats or tax rate changes. One potential reform would be to consolidate all ambiguous assets into a single Box, removing discretion over their classification. This would curtail opportunities for arbitrage between Boxes and ensure that tax reforms—such as adjustments to marginal rates or wealth taxes—are not diluted by intra-return shifting. The likely consequence would be improved policy targeting and a more transparent relationship between tax liability and asset holdings.

Audit Announcements as Self-Financing Compliance Tools

Targeted audit announcements may be self-funding if the long-term increase in declarations,

driven by taxpayers learning about reporting obligations or heightened perceived enforcement, exceeds the upfront costs of the announcement, the associated audits, and any subsequent decline in declarations among dishonest taxpayers (‘crater’ effects).

Dividend Tax Cuts as Liquidity Instruments

The 2007 temporary dividend tax cut produced an immediate revenue windfall of approximately €2.8 billion, driven by large one-time payouts. This suggests that temporary rate reductions can be used to unlock liquidity from retained corporate earnings in fiscally constrained years. However, frequent use of this instrument risks a behavioural offset: firms may delay payouts in anticipation of future cuts, possibly reducing average annual revenue. To avoid this dynamic, such rate changes should be clearly communicated as one-off measures, ideally accompanied by commitments not to repeat them in the short term.

8 Conclusion

This paper illustrates how ambiguity in asset classifications and flexibility in tax reporting allow taxpayers to engage in strategic forms of tax evasion and avoidance. Using the 2005 and 2007 auditing announcements and a temporary dividend tax cut in the Netherlands, the analysis shows how taxpayers adapt to audit threats by shifting their declarations strategically. Specifically, the results reveal that targeted audits can prompt taxpayers to adjust declarations across different sections of their tax returns to exploit lower tax rates and minimise exposure.

These findings underscore how ambiguous distinctions between tax categories enable shifting behaviour, and how taxpayers exploit this flexibility in response to both audit threats and changes in tax incentives. While streamlining reporting requirements may help, a more targeted reform would involve clearly delineating, or bundling into a single category, assets whose classification currently depends on taxpayer discretion.

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APPENDIX

A Yearly announcements

In our analysis we must exclude any intervening effects from other announcements between 2002-2008. This is likely to hold. The 2002 announcement concerned uncommon expenditures and the 2003-2004 announcements concerned pension annuity payments, neither of which overlap with the topics we study. The 2006 announcement concerned a specific type of deductible on mortgage debt for people who sell a house and buy a new one within the same fiscal year. This announcement may overlap slightly with some of the property sub-items targeted in the 2005 and 2007 announcements but, if unaccounted for, would likely lead to an underestimate of our misreporting effects. The influence of the 2006 announcement is unlikely to be large since it affects only a small fraction of house buyers in 2006. The 2008 announcement concerned charitable donations which are separate from the topics we study in Box 1 and Box 3.

Table A1: Belastingdienst Annual Audit Focus Topics, 2002–2010

Year	Dutch topic (official/Belastingdienst wording)	English explanation
2002	Buitengewone uitgaven	Deductions for specific personal expenses (e.g., health, disability), with audits focusing on correct eligibility and documentation.
2003	Lijfrenteaf trek	Pension annuity deductions, especially correct use of available deduction space and payment rules.
2004	Lijfrenteaf trek	Continued focus on correct computation and documentation of pension annuity deductions.
2005	Bijverdiensten	Side earnings (e.g., freelance, artist, caregiver work); audits focused on under-reporting or misclassification.
2006	Bijleenregeling (eigen woning)	Mortgage interest deductions when buying/selling a home in the same year; audits targeted correct offsetting of proceeds.
2007	Box 3 (sparen en beleggen)	Income from savings and investments; audits focused on correct asset and debt valuations as of 1 January.
2008	Giftenaftrek	Charitable donation deductions, focusing on ANBI eligibility and limits.
2009	Kinderalimentatie en bijdrage voor kinderen	Deductions for child support or financial contributions; audits checked eligibility and records.
2010	Tijdig indienen van aangifte	Timely tax return submission; focus was on compliance with deadlines and related penalties.

B Descriptive Statistics

Table A2: Declarations by individual characteristics

	Repr.	Targ.	Link.
All	100%	100%	100%
Gender			
Female	50.8%	53.8%	22.3%
Male	49.2%	46.2%	77.7%
Fiscal Partner			
Single	67.0%	29.5%	18.1%
Partner	33.0%	70.5%	81.9%
Migrant			
Dutch	97.0%	96.5%	97.5%
Foreign	3.0%	3.5%	2.5%
Children			
0	37.4%	26.4%	21.2%
1–2	47.5%	51.4%	52.9%
3+	15.1%	22.2%	25.9%
Age			
<30	7.2%	10.8%	3%
30–50	39.2%	46.3%	49.9%
50–65	32.2%	33.3%	37.1%
65+	21.4%	9.6%	10%
<i>N</i> ind.	65,005	33,666	50,147
<i>N</i> obs.	455,035	235,662	351,029

Column 1 is representative sample, column 2 is targeted *income from side earnings* sample and column 3 is linked *profits from assets made available* sample.

Table A3: Declarations by employment characteristics

	Repr.	Targ.	Link.
All	100%	100%	100%
Income			
<0	1.7%	2.0%	2.8%
0–30k	49.7%	57.3%	28.4%
30–60k	27.1%	22.7%	35.1%
60–120k	7.9%	7.7%	22.7%
120k+	2.2%	2.0%	10.9%
Employment			
entrepr.	10.6%	9.4%	14.2%
non-entrepr.	89.4%	90.6%	85.8%
Tax prof.			
no	60.2%	65.5%	17.0%
yes	39.8%	34.5%	83.0%
Sector			
agricul.	1.0%	1.2%	1.9%
industry	4.7%	3.2%	4.0%
instal.	2.5%	1.3%	3.8%
wholesale	3.2%	2.0%	5.5%
retail	3.7%	4.1%	6.7%
trans-comm	1.9%	1.4%	1.9%
bank-fin.	13.5%	12.5%	45.6%
other serv.	14.8%	22.5%	5.7%
unempl.	54.3%	51.1%	24.8%
none	0.4%	0.7%	0.1%
<i>N</i> ind.	65,005	33,666	50,147
<i>N</i> obs.	455,035	235,662	351,029

The entrepreneur category includes people who registered as independent without personnel' or small firms. Small firms are those with fewer than 5 employees. Classification as entrepreneur follows tax-administration definitions of zelfstandig ondernemerschap and eligibility for self-employment reporting, and does not preclude wage employment. The *Employment* category omits observations in 2002 since these have no information concerning entrepreneur status. Job sector categories: agricul.=agriculture and fisheries; industry=industry and mineral extraction; instal=construction, installation and utilities; wholesale=wholesale and intermediate trade; retail=retail, catering and repair; trans-comm.=transport, storage and communication; bank-fin.=banking, insurance and business services. Column 1 is representative sample, column 2 is targeted *income from side earnings* sample and column 3 is linked *profits from assets made available* sample.

Table A4: Summary statistics (in €)

	Repr. sample avg. frac.		Targ. sample avg. frac.		Link. sample avg. frac.	
Total	30,649	88.7%	27,502	91.8%	70,733	97.2%
	(98,811)		(59,143)		(228,872)	
Box 1 total	26,252	86.4%	24,893	91.2%	46,273	96.4%
	(35,662)		(37,688)		(73,941)	
side earn.	286	4.9%	2,896	58.3%	667	6.8%
	(4,211)		(10,564)		(16,562)	
other profits	97	2.0%	40	1.5%	2,711	62.3%
	(3,366)		(3,652)		(41,576)	
Box 2 total	1,777	1.5%	936	0.8%	15,466	9.7%
	(87,668)		(31,988)		(202,495)	
Box 3 total	2,620	51.5%	1,672	25.3%	8,994	53.1%
	(14,074)		(23,706)		(41,434)	
other property	19,820	8.6%	13,716	5.5%	16,349	28.2%
	(231,522)		(383,611)		(1,188,623)	
other assets	1,888	2.1%	1,075	1.4%	8,825	4.9%
	(35,391)		(24,726)		(237,280)	
debts	15,603	18.8%	15,016	12.4%	135,192	47.2%
	(186,956)		(778,515)		(1,081,514)	
shares, bonds, etc.	27,340	27.9%	23,207	16.3%	90,842	35.1%
	(203,514)		(1,156,649)		(696,930)	
savings	40,487	55.1%	21,441	28.5%	77,903	61.5%
	(127,129)		(111,740)		(378,735)	
other claims & cash	5,231	7.8%	2,929	5.1%	25,014	16.9%
	(118,148)		(38,844)		(280,144)	
2nd home	4,445	3.2%	2,922	2.1%	16,408	6.7%
	(46,077)		(31,319)		(172,029)	
capital insurance	398	2.6%	340	1.5%	2,081	5%
	(10,992)		(12,901)		(51,963)	
benefits claims	77	0.2%	51	0.1%	335	0.3%
	(4,405)		(4,522)		(39,936)	

Standard deviations in parenthesis. *frac.* represents the average fraction of taxpayers in each sample declaring the specific tax category over years 2002–2008. Columns 1–2 are representative sample, columns 3–4 are targeted *income from side earnings* sample, and columns 5–6 are linked *profits from assets made available* sample.

B.1 Additional Descriptive Figures

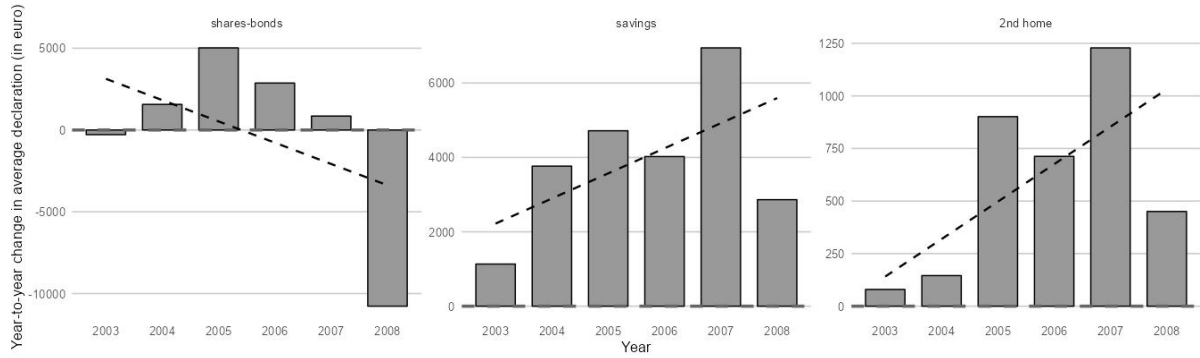


Figure A1: Yearly changes in average declarations, 2002–2008

Notes: This figure is constructed from Dutch administrative longitudinal tax records covering a representative sample of 65,005 individuals who filed taxes in each year from 2002 to 2008. The plotted values show annual averages of the first difference in declared amounts for each asset category presented in the paper. The dashed line overlays the baseline linear trend estimated in our main specification, illustrating the underlying model-implied trajectory against observed changes.

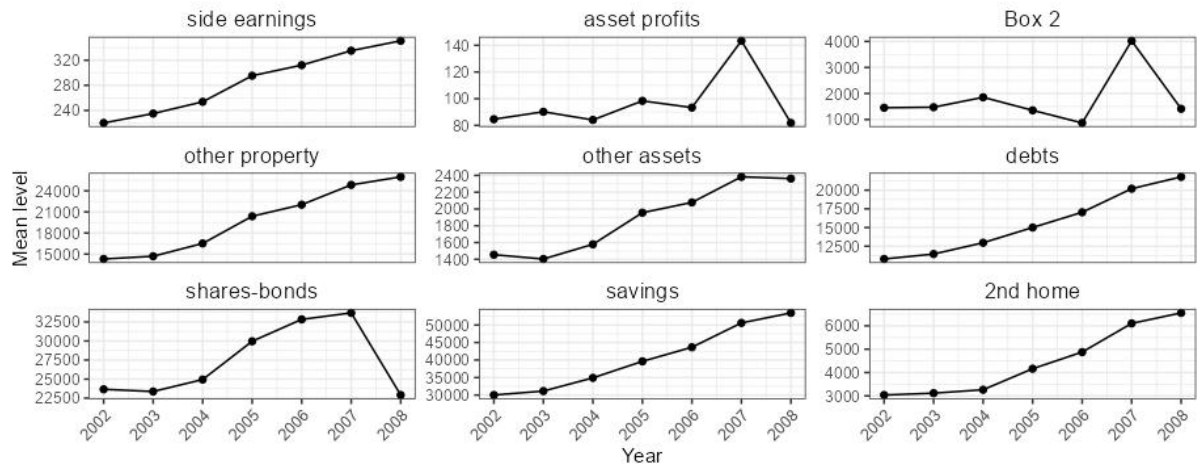


Figure A2: Levels of declared amounts, 2002–2008

Notes: This figure is constructed from Dutch administrative longitudinal tax records covering a representative sample of 65,005 individuals who filed taxes in each year from 2002 to 2008. The plotted values show average declared levels (in euros) for each asset category presented in the paper. For each category, the annual mean value is plotted over time to illustrate level dynamics.

B.2 Direct Effects on Side Earnings in Box 1

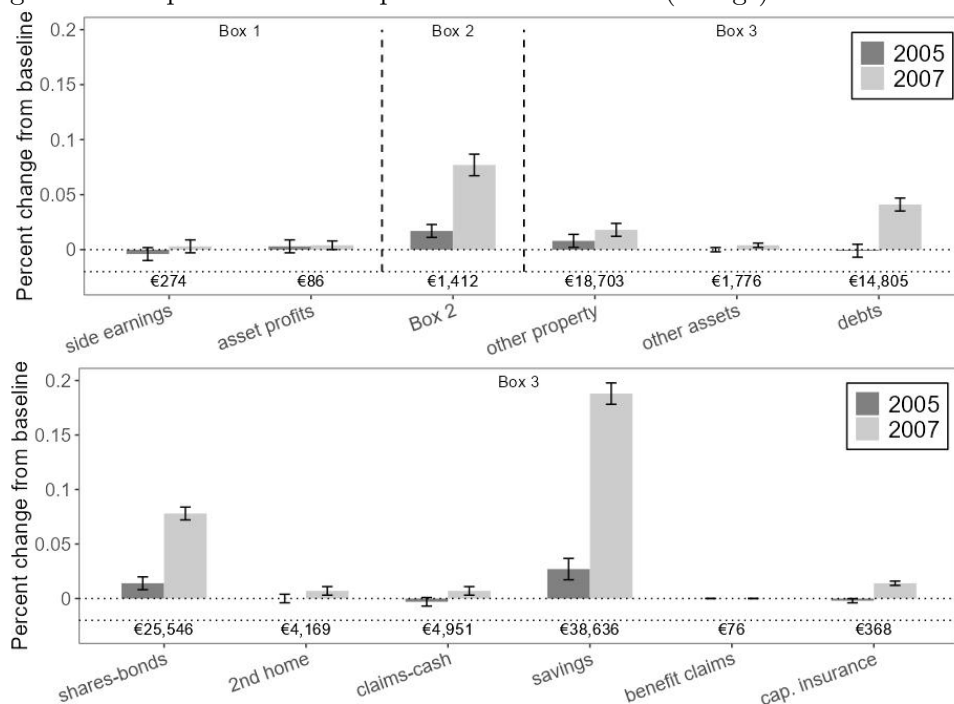
Table A5: Bump or lasting changes in declarations? Outcome: *Side earnings*, conditional on declaration in $t-1$

	2003	2004	2005	2006	2007	2008
Effect	25.721 (35.632) [0.470]	-379.387*** (110.022) [0.001]	-191.348*** (42.265) [0.000]	-371.068*** (93.771) [0.000]	-160.941*** (35.806) [0.000]	-1604.068*** (85.718) [0.000]
Mean	2895.608	2895.608	2895.608	2895.608	2895.608	2895.608
Observations	235,662	235,662	235,662	235,662	235,662	235,662
Individuals	33,666	33,666	33,666	33,666	33,666	33,666

Notes: This table estimates the change in *income from side earnings* declarations at year t , conditional on having declared non-null *income from side earnings* in year $t-1$. The table evaluates the additional effect of the letters sent to taxpayers who declared *income from side earnings* in 2004. The results show that *income from side earnings* is typically transitory: taxpayers who declare it in $t-1$ are less likely to do so in t . Coefficients with standard errors are in parentheses and p-values in brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

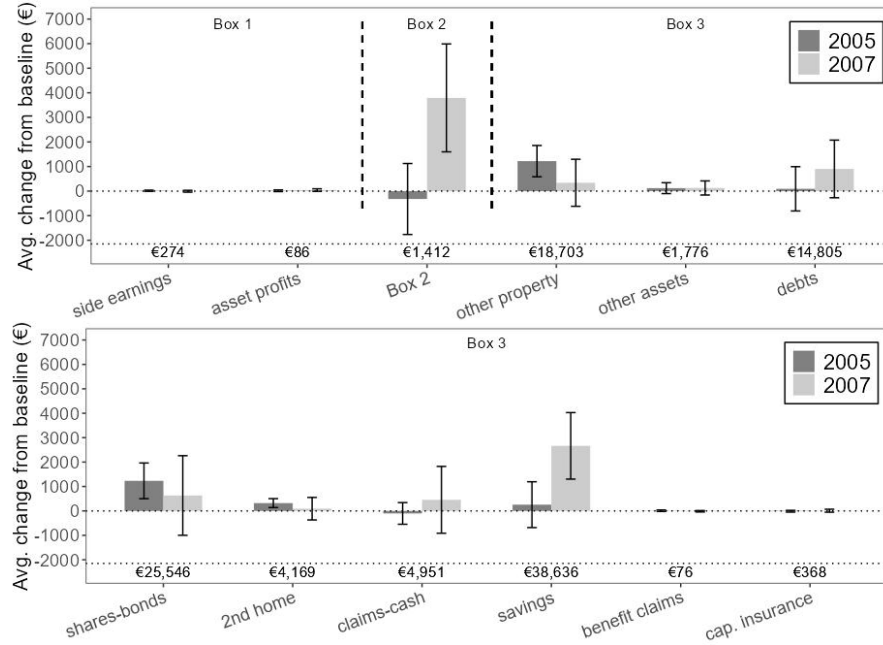
B.3 Additional Results

Figure A3: Representative sample declaration effects (in logs) in 2005 and 2007



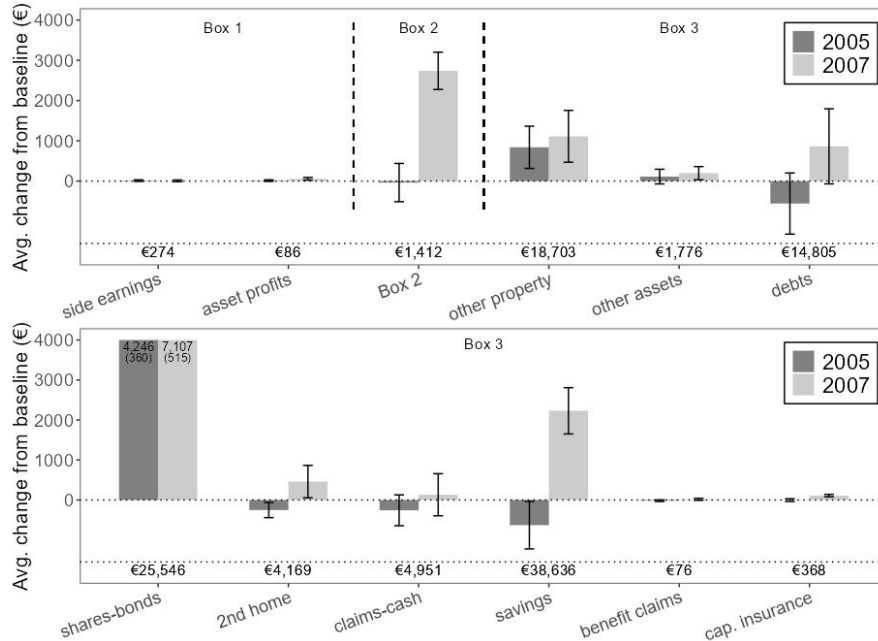
Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level regressions of first differences in declared amounts on year indicators, as specified in Equation 1, using log-transformed outcomes. Declarations below €1,000 are imputed to €1,000 to avoid excessive leverage from very small values. Units are log euros per taxpayer. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each log outcome over 2002–2008, excluding 2005 and 2007, for the balanced panel of 65,005 individuals.

Figure A4: Representative sample declaration effects (in levels) in 2005 and 2007, third degree polynomial in time



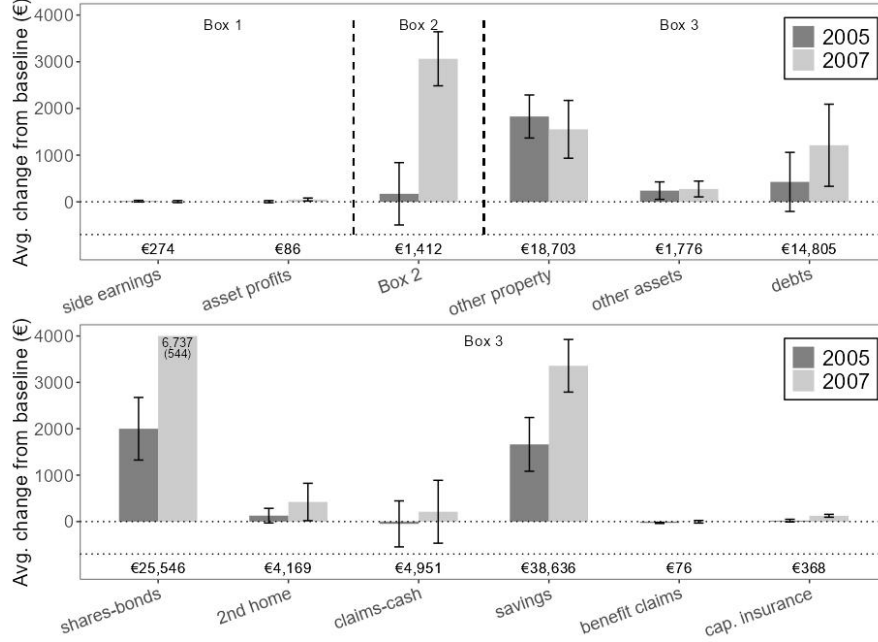
Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level regressions of first differences in declared amounts on year indicators, as specified in Equation 1, using a third-degree polynomial in time in place of the linear time trend. Units are euros per taxpayer. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each outcome over 2002–2008, excluding 2005 and 2007, for the balanced panel of 65,005 individuals.

Figure A5: Representative sample declaration effects in 2005 and 2007, within estimator



Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level within (fixed-effects) regressions of declared amounts on year indicators and a linear time trend with individual fixed effects. Units are euros per taxpayer. Bars show the estimated change in average declarations relative to the baseline time trend and individual fixed effects. For the *Box 2* estimate in 2007, the bar is truncated for presentation purposes; the exact estimate and its standard error are printed on the truncated bar in the figure. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each outcome over 2002–2008, excluding 2005 and 2007, for the balanced panel of 65,005 individuals.

Figure A6: Representative sample declaration effects in 2005 and 2007, baseline trend estimated on pre-2005 data



Notes: This figure plots coefficient estimates and 95% confidence intervals of β_{2005} and β_{2007} obtained from separate taxpayer-level regressions of first differences in declared amounts on year indicators, using a linear time trend estimated only on pre-2005 observations. Units are euros per taxpayer. Bars show the estimated change in average declarations relative to this pre-2005 baseline trend and individual fixed effects. For the *Box 2* estimate in 2007, the bar is truncated for presentation purposes; the exact estimate and its standard error are printed on the truncated bar in the figure. Standard errors are clustered at the individual level. The means displayed at the bottom of the panels correspond to taxpayer-level averages of each outcome over 2002–2008, excluding 2005 and 2007, for the balanced panel of 65,005 individuals.

Table A6: Representative sample: Fraction of Positive, Zero, and Negative Instances by Category and Year (2002–2008)

Category		2002	2003	2004	2005	2006	2007	2008
Side Earn.	Positive	0.039	0.040	0.042	0.046	0.048	0.051	0.052
	Zero	0.959	0.957	0.955	0.950	0.949	0.945	0.944
	Negative	0.002	0.003	0.003	0.003	0.003	0.003	0.004
Asset Profits	Positive	0.017	0.018	0.018	0.018	0.019	0.020	0.019
	Zero	0.981	0.980	0.979	0.979	0.979	0.978	0.979
	Negative	0.002	0.002	0.002	0.002	0.002	0.002	0.002
Box 2	Positive	0.009	0.009	0.010	0.009	0.009	0.028	0.012
	Zero	0.990	0.990	0.988	0.990	0.987	0.969	0.983
	Negative	0.001	0.001	0.001	0.001	0.003	0.004	0.004
Other Property	Positive	0.081	0.070	0.084	0.087	0.088	0.094	0.092
	Zero	0.919	0.930	0.916	0.913	0.911	0.906	0.908
Other Assets	Positive	0.020	0.018	0.019	0.021	0.021	0.024	0.024
	Zero	0.980	0.982	0.981	0.979	0.978	0.976	0.976
Debts	Positive	0.124	0.116	0.130	0.148	0.225	0.297	0.279
	Zero	0.876	0.884	0.870	0.853	0.775	0.703	0.721
Shares-Bonds	Positive	0.295	0.252	0.268	0.284	0.290	0.297	0.270
	Zero	0.705	0.748	0.732	0.716	0.710	0.703	0.730
	Negative	0.000	0.000	0.000	0.000	0.000	0.000	0.000
Savings	Positive	0.488	0.433	0.491	0.546	0.592	0.670	0.628
	Zero	0.511	0.565	0.508	0.453	0.406	0.329	0.370
Tax Advisor	Yes	0.203	0.360	0.396	0.420	0.451	0.476	0.477
	No	0.797	0.640	0.604	0.580	0.549	0.524	0.523
Side Earn.	Positive	0.000	0.131	0.121	0.123	0.123	0.122	0.122
	Zero	1.000	0.869	0.879	0.877	0.877	0.878	0.878